

DATA DICTIONARY — WILDFIRE PREDICTION DATASET

Project	Wildfire Prediction Pipeline (Big Data & AI course project)
Source	Kaggle - Wildfire Prediction by Vijayaragul VR
File	final_dataset.csv
Rows	118,858
Columns	17
Missing values	0
Duplicate rows	0

Data origin

- frp - NASA FIRMS VIIRS SCC (Suomi NPP satellite)
- All meteorological features - Open-Meteo (1-hour aggregation window)
- lat, lon, daynight_N, occured - not explicitly documented by the Kaggle publisher

Field Definitions

Each field is listed with its name, data type, unit (where applicable), role in the pipeline, and value meanings. Categorical fields enumerate their possible values.

DAYNIGHT_N float64 (binary)
Time-of-day indicator for the observation
Role: predictor feature
 0 .Nighttime
 1 .Daytime

LAT float64
Latitude coordinate of the observation point
Unit: degrees (°)
Role: predictor feature (Models 1 & 2 only)
Range: -51.78 to 71.23

LON float64
Longitude coordinate of the observation point
Unit: degrees (°)
Role: predictor feature (Models 1 & 2 only)
Range: -158.04 to 179.58

FIRE_WEATHER_INDEX float64
Composite fire-danger index derived from multiple weather factors
(temperature, humidity, wind, precipitation)
Unit: dimensionless
Role: predictor feature
Range: -16.92 to 211.63

PRESSURE_MEAN float64
Average atmospheric surface pressure over the 1-hour window
Unit: hPa
Role: predictor feature
Range: 589.64 to 1035.95

WIND_DIRECTION_MEAN float64
Average wind direction over the 1-hour window, clockwise from north
Unit: degrees (0°-360°)
Role: predictor feature

WIND_DIRECTION_STD float64
Standard deviation of wind direction over the 1-hour window
Unit: degrees
Role: predictor feature
Notes: High values indicate gusty or shifting winds

SOLAR_RADIATION_MEAN float64
Average shortwave surface solar radiation during the 1-hour window
Unit: W/m²
Role: predictor feature

DEWPOINT_MEAN float64
Average dew-point temperature over the 1-hour window
Unit: °C
Role: predictor feature
Notes: Direct measure of atmospheric moisture – lower dew points favor fire

CLOUD_COVER_MEAN float64
Average total cloud-cover fraction over the 1-hour window
Unit: % (0-100)
Role: predictor feature

EVAPOTRANSPIRATION_TOTAL float64
Total evapotranspiration (land surface + plants) over the 1-hour window
Unit: mm
Role: predictor feature
Notes: Strongly correlated with fire_weather_index (r = +0.84)

HUMIDITY_MIN float64
Minimum relative humidity during the 1-hour window
Unit: % (0-100)
Role: predictor feature
Notes: Lower values strongly associated with fire occurrence

TEMP_MEAN float64
Average air temperature at 2m above ground during the 1-hour window
Unit: °C
Role: predictor feature
Range: -49.05 to 41.55

TEMP_RANGE float64
Temperature range (max – min) during the 1-hour window
Unit: °C
Role: predictor feature

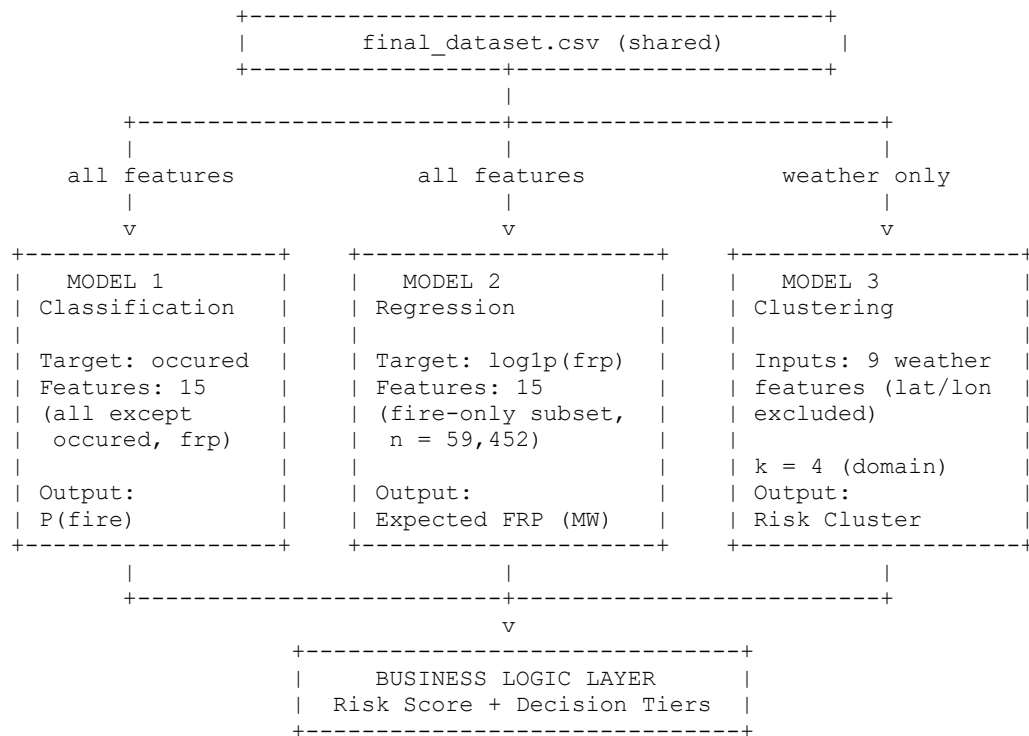
WIND_SPEED_MAX float64
Maximum wind speed at 10m above ground during the 1-hour window
Unit: km/h
Role: predictor feature

OCCURED float64 (binary)
Fire occurrence label
Role: TARGET – Model 1 (Classification)
Notes: Dataset is perfectly balanced (50.0% / 50.0%) due to

deliberate sampling by the Kaggle publisher
0 .No fire detected
1 .Fire detected

FRP float64
Fire Radiative Power – rate of radiant heat output from a fire,
measured by the VIIRS thermal sensor aboard Suomi NPP
Unit: MW (megawatts)
Role: TARGET – Model 2 (Regression)
Range: 0.00 to 1056.15
Notes: Heavily right-skewed (skew = 6.18); log-transform recommended

Role in the Three-Model Pipeline



Feature Groupings

- Spatial (2): lat, lon
- Temporal (1): daynight_N
- Thermal (3): temp_mean, temp_range, dewpoint_mean
- Moisture (2): humidity_min, evapotranspiration_total
- Wind (3): wind_direction_mean, wind_direction_std, wind_speed_max
- Radiation & cloud (2): solar_radiation_mean, cloud_cover_mean
- Pressure (1): pressure_mean
- Composite index (1): fire_weather_index
- Targets (2): occurred, frp